

Lecture 4: Inference for Subpopulation Proportions and Differences

August 23, 2023

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- ▶ We saw that for large enough sample sizes, our sample proportion/mean was likely to be close to the population proportion/mean.
- ▶ We also demonstrated that we could predict how close we might be by using confidence intervals.
- ▶ We explored a bootstrapping approach and a plug-in approach to producing confidence intervals.

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- ▶ Even when the overall sample size is fixed (625 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.

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- ▶ Returning to the turnout example, we will consider Black turnout and the difference between Black and non-Black turnout in the 2020 Georgia presidential election.
- ▶ Even when the overall sample size is fixed (625 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.
- ▶ This complexity will not complicate our analysis much but it does complicate the justification for the analysis.

2020 Presidential Election Turnout in Georgia

We return to the turnout application, but we are now interested in Black turnout and Black turnout in comparison to non-Black turnout. Recall that ...

- ▶ based on a survey of 625 registered voters a week before the election, we got a turnout rate of .68,
- ▶ after the election, we found that the turnout rate among all registered voters was .70

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- ▶ in our survey of 625 registered voters, 184 were Black with a turnout rate of .71 and 441 were non-Black with a turnout rate of .67.

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Suppose we also want to predict Black turnout and the difference between Black and non-Black turnout.

- ▶ in our survey of 625 registered voters, 184 were Black with a turnout rate of .71 and 441 were non-Black with a turnout rate of .67.
- ▶ after the election, we found that the turnout rate among the 30% of registered voters who were black was .74 and among the 70% of non-Black registered voters was .68.

Four Questions

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Four Questions

1. .71 is close to the Black turnout rate of .74. Did we get lucky?
2. Could we have predicted how close we would get to .74 before the election?
3. The survey turnout gap of .04 is close to the population turnout gap of .06. Did we get lucky?
4. Could we have predicted how close we would get to a .06 turnout gap before the election?

Sampling Distribution for Black Turnout Proportion

Let the variable X take the value 1 if Black and 0 if non-Black

i	1	2	...	625		
	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\bar{Y}_{625}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$
	1,0	1,1	...	0,0	.68	.69
$f(\cdot)$			...		$N(.7, \frac{.7 \cdot (1-.7)}{n})$	
$E(\cdot)$			...		.7	
$V(\cdot)$			...		$.7 \cdot (1 - .7)$	

Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV, so we use the general CLT approach.

Sampling Distribution for Black Turnout Proportion

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s	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\bar{Y}_{625}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$
1	1,0	1,1	...	0,0	.68	.69
2	0,0	1,0	...	1,1	.70	.72
$f(\cdot)$			...		$N(.7, \frac{.7 \cdot (1 - .7)}{n})$	
$E(\cdot)$			...		.7	
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1	1,0	1,1	...	0,0	.68	.69
2	0,0	1,0	...	1,1	.70	.72
3	1,0	1,1	...	0,1	.75	.76
4	1,1	1,0	...	0,0	.73	.73
5	0,0	0,0	...	1,1	.69	.71
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	1,0	0,1	...	1,1	.74	.75
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$			...		$N(.7, \frac{.7 \cdot (1-.7)}{n})$	
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$V(\cdot)$			...		$.7 \cdot (1 - .7)$	

Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV, so we use the general CLT approach.

What can we learn?

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	?
$E(\cdot)$	μ	μ	...	μ	?
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

- ▶ Although we can't observe the population or the sampling distribution, we can mathematically derive certain facts about the sampling distribution.
- ▶ We will focus on the three red question marks.

The Type of Distribution

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f.(\cdot)$	?	?	...	?	$N(?, ?)$
$E.(\cdot)$	μ	μ	...	μ	?
$V.(\cdot)$	σ^2	σ^2	...	σ^2	?

Property #1: *The sampling distribution of the sample mean is approximately normal when n is large.*

- ▶ The approximate normality result comes from the central limit theorem (CLT).
- ▶ We know that $\bar{Y}_{2258} \sim^{approx} N(?, ?)$, but still don't know the mean or the variance of this distribution.

What is the $E[\bar{Y}_n]$?

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	$N(\textcolor{red}{?}, ?)$
$E(\cdot)$	μ	μ	...	μ	?
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

What is the $E[\bar{Y}_n]$?

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f.(.)$	?	?	...	?	$N(?, ?)$
$E.(.)$	μ	μ	...	μ	?
$V.(.)$	σ^2	σ^2	...	σ^2	?

Property #1: *The sample mean is unbiased for the population mean.* The expected value of the sample mean, $E[\bar{Y}] = \mu$.

Proof:

$$\begin{aligned} E[\bar{Y}] &= E\left[\frac{1}{n}(Y_1 + \dots + Y_n)\right] \\ &= \\ &= \\ &= \\ &= \\ &= \mu. \end{aligned}$$

Sample mean is unbiased for the population mean

i	1	2	...	2258	
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1	37,006	108,811	...	49,001	135,406
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3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	$N(\mu, ?)$
$E(\cdot)$	μ	μ	...	μ	μ
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

What is the $V[\bar{Y}_n]$?

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	:	:	...	:	:
5	:	:	...	:	:
:	:	:	:	:	:
1 mil	?	?	...	?	?
:	:	:	:	:	:
:	:	:	:	:	:
$f(\cdot)$	?	?	...	?	$N(\mu, ?)$
$E(\cdot)$	μ	μ	...	μ	μ
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

Property #2: The *sampling variance* of the sample mean,

$$V[\bar{Y}] = \frac{\sigma^2}{n}$$

Proof:

$$\begin{aligned} V[\bar{Y}] &= V \left[\left(\frac{1}{n} Y_1 + \dots + \frac{1}{n} Y_n \right) \right] \\ &= \\ &= \\ &= \\ &= \\ &= \frac{\sigma^2}{n}. \end{aligned}$$

How much does $\bar{Y}_{2258}$ vary from sample to sample?

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	:	:	...	:	:
5	:	:	...	:	:
:	:	:	:	:	:
1 mil	?	?	...	?	?
:	:	:	:	:	:
:	:	:	:	:	:
$f.(.)$	?	?	...	?	$N(\mu, \frac{\sigma^2}{n})$
$E.(.)$	μ	μ	...	μ	μ
$V.(.)$	σ^2	σ^2	...	σ^2	$\frac{\sigma^2}{n}$

Facts about the sampling distribution of $\bar{Y}_n$

What we know about the sampling distribution of $\bar{Y}_n$:

1. Approximately normally distributed
2. Centered at μ .
3. Variance of $\frac{\sigma^2}{n}$

These facts imply that the middle 95% of the distribution lies approximately between $\mu - 1.96 \cdot \frac{\sigma}{\sqrt{n}}$ and $\mu + 1.96 \cdot \frac{\sigma}{\sqrt{n}}$. As long as $\sqrt{n}$ is big in relation to σ , $\bar{Y}_n$ will likely be close to μ .

We still don't know:

1. The value of μ .
2. The value of σ .

Two Questions

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Two Questions

1. Is \$135,406 close to the population mean μ ? Yes!
2. Can say something about how close? Let's use a confidence interval. We will look at two methods: bootstrapping and plug-in.

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However, we can bootstrap the sampling distribution by taking repeated random samples *with replacement* of size n from the sample of size n .

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y_i	37,006	108,811	...	49,001	135,406

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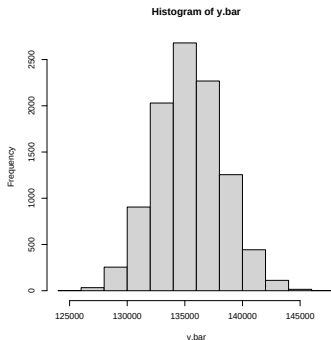
i	1	2	...	2258	$\bar{y}_{2258}$
y_i	37,006	108,811	...	49,001	135,406

With the data stored in the vector y , each mean of the bootstrapped sampling distribution can be generated with

```
mean(sample(y, size=2258, replace=TRUE))
```

Bootstrapped CIs for a Population Mean

```
B=10000  
y.bar = rep(NA,B)  
for(b in 1:B){y.bar[b]= mean(sample(y,size=n,replace=T))}  
hist(y.bar)
```



```
> quantile(y.bar,probs=c(.025,.975)) ## bootstrapped CI  
      2.5%      97.5%  
129783.2 141169.0
```

Bootstrapped Sampling Distribution vs Sampling Distribution

1. CLT says the sampling distribution is approximately normal.
2. We see that the bootstrapped distribution is approximately normal.
3. Remaining question is whether the bootstrapped CI has approximately the same size as the middle 95% of the sampling distribution.

Let's table the remaining question for now while we look at another approach to CIs.

Plug-in CI

- ▶ Another way to get a 95% CI is to use the fact that the middle 95% of the distribution lies approximately between $\mu - 1.96 \cdot \frac{\sigma}{\sqrt{n}}$ and $\mu + 1.96 \cdot \frac{\sigma}{\sqrt{n}}$.
- ▶ If we denote the sample standard deviation with s , then we plug in sample values into the above statement:
 $\bar{y} - 1.96 \cdot \frac{s}{\sqrt{n}}$ and $\bar{y} + 1.96 \cdot \frac{s}{\sqrt{n}}$
- ▶ 129,720 and 141,091 (note how close these are to the bootstrapped CI)

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Why does this work?

The facts we've derived about the sampling distribution of the sample mean tell us that

$$P(\mu - 1.96 \cdot \frac{\sigma}{\sqrt{n}} \leq \bar{Y} \leq \mu + 1.96 \cdot \frac{\sigma}{\sqrt{n}}) \approx .95$$

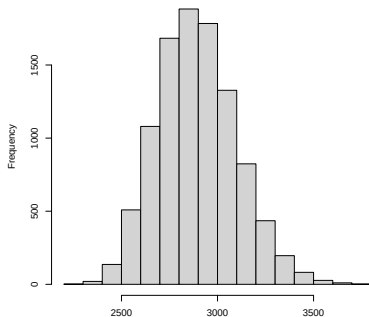
Some algebra (left as an exercise) shows that this implies

$$P(\bar{Y} - 1.96 \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{Y} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}) \approx .95$$

Remaining question is whether $\frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$ (note how similar this is to the remaining bootstrapped CI question).

Partially Addressing Remaining Question

- ▶ $\frac{\sigma}{\sqrt{n}}$ is called a standard error because it is the standard deviation for an estimator ($\bar{Y}$) of a parameter (μ).
- ▶ $\frac{s}{\sqrt{n}}$ is called an estimated standard error. We will bootstrap this estimated standard error to get a sense of its uncertainty (note the scale of this uncertainty).



CI Wrap Up

We got the following 95% CIs for the population income mean:

- ▶ Bootstrap: \$129,783 to \$141,169
- ▶ Plug-in: \$129,720 to \$141,091

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- ▶ Bootstrap: \$129,783 to \$141,169
- ▶ Plug-in: \$129,720 to \$141,091

From the previous slide, we see the estimated standard error is very unlikely to be off by more than \$1,000, and therefore the bounds are very unlikely to be off by more than \$2,000.

We are at least 95% confident that the true population mean is roughly between \$128,000 and \$143,000, despite observing only a tiny fraction of the population.

Two Questions

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2. Can say something about how close? Yes! Likely within \$7,500 or so.

Summary

...